

GLM

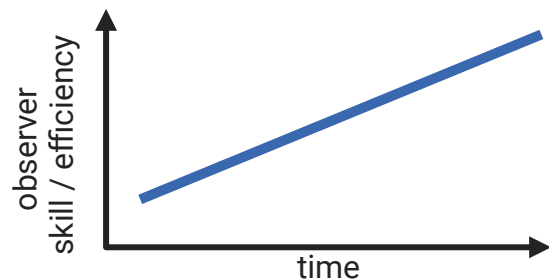
**combined abundance/
detection model:**
 $y_i \sim \text{Poisson}(\lambda_i)$

detection confounds:

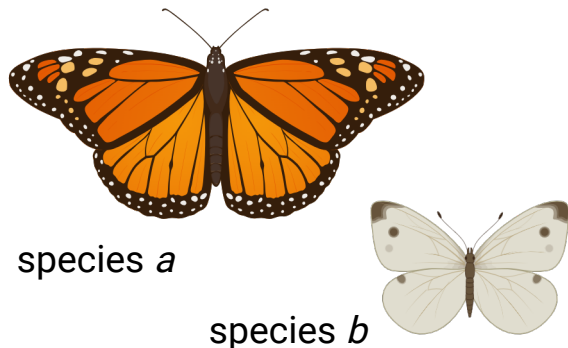
spatial



temporal



taxonomic



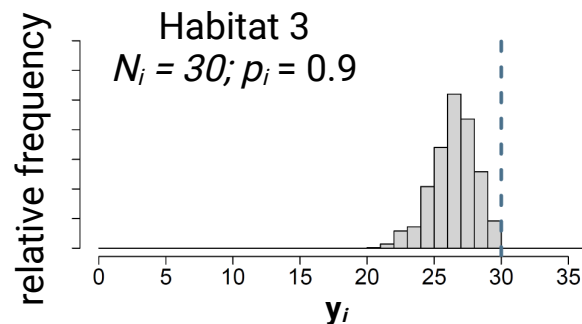
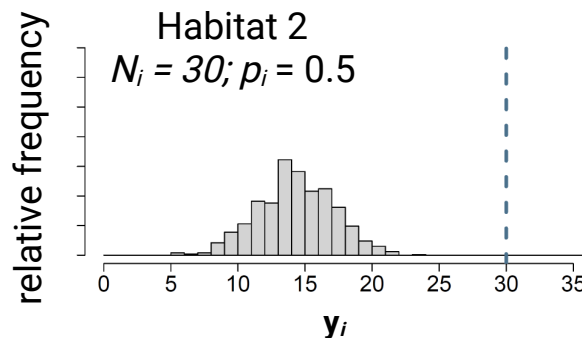
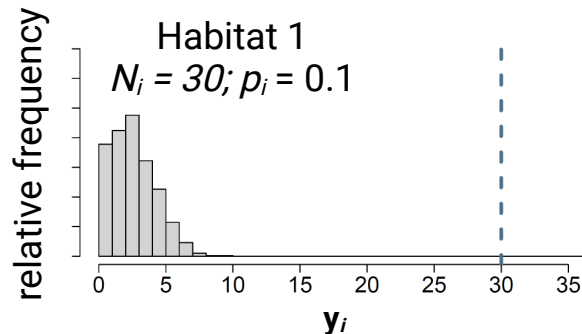
Binomial N-mixture

abundance model:

$$N_i \sim \text{Poisson}(\lambda_i)$$

detection model:

$$y_{ij} \sim \text{binomial}(N_i, p_i)$$



Multinomial N-mixture

abundance model:

$$N_i \sim \text{Poisson}(\lambda_i)$$

detection model:

$$n_i \sim \text{binomial}(N_i, 1 - \pi_0);$$

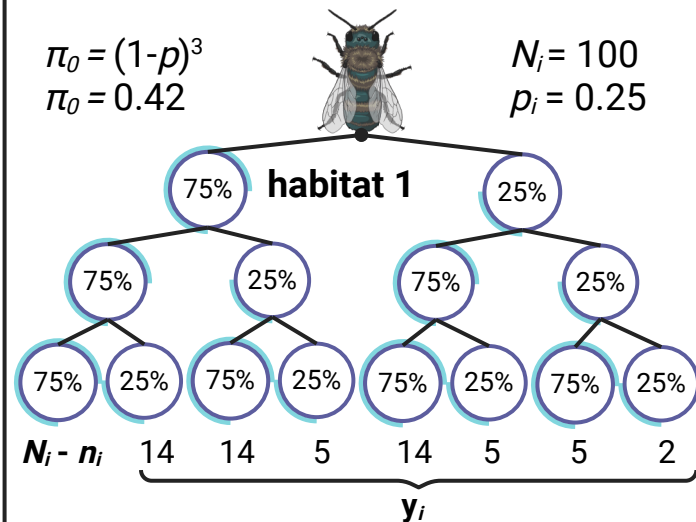
$$y_i | n_i \sim \text{multinomial}(n_i, \pi_i^c)$$

$$\pi_0 = (1-p)^3$$

$$\pi_0 = 0.42$$

$$N_i = 100$$

$$p_i = 0.25$$



$$\pi_0 = (1-p)^3$$

$$\pi_0 = 0.125$$

$$N_i = 100$$

$$p_i = 0.5$$

